TRANSPONDER LANDING SYSTEM (TLS) CALIBRATION MANUAL

ADVANCED NAVIGATION & POSITIONING CORPORATION

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NOTICE

Advanced Navigation and Positioning Corporation has attempted to describe in this manual all areas of possible danger to personnel in connection with the use of this equipment. This document refers to United States government and professional technical standards: Consult government standards prevailing in the region of installation to assure compliance with local requirements.

Personnel should use caution when checking, operating, and servicing this equipment, especially when power is on. Like all electronic equipment, care should be taken to avoid electrical shock in all circuits where substantial currents or voltages may be present, through design or short circuit. Caution should be observed also in lifting and hoisting equipment especially regarding large structures during installation.

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ENGINEERING CHANGE ORDER RECORD
TRANSPONDER LANDING SYSTEM (TLS) CALIBRATION MANUAL

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Revision	Issue Date	Revision Description	ECN #
A	11/30/12	TLS Manual Division	1916
B	2014Dec09	Update per PFR 3292	1934
C	2018Apr11	Update customer support numbers and sections for theodolite setup and use	02046
D	2019Oct02	Added helicopter calibration procedure for low visibility	ECA-00148
E	2020Dec22	Removed requirement to place theodolite at Loc point for verification, other edits for clarity	ECA-00225

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SECTION 1: INTRODUCTION

1.1 Document Overview

This manual describes the procedural steps necessary to perform calibration and validation of the Transponder Landing System (TLS). Section 2 describes the assemblies that need to be readied and in place prior to the calibration, the tools (software and hardware) to use to calibrate and procedures for calibrating the system. Section 3 describes the tools and procedures performed during flight check of the system.

1.2 ANPC TLS Manuals

The ANPC TLS is accompanied by a group of four primary manuals:

- Document # 020-00071, TLS Calibration Manual
- Document # 020-00072, TLS Operations Manual
- Document # 020-00073, TLS Installation Manual
- Document # 020-00074, TLS Maintenance Manual

The ANPC TTLS includes two additional manuals:

- Document 020-00076 Container System Deployment
- Document 020-00077 TTLS Packing Instructions

SECTION 2: CALIBRATION AND VALIDATION

NOTE

Before calibration, it is important to verify that vegetation and/or snow has been cleared within tolerances. See Maintenance Procedure M-3 "Maintenance of Critical Areas" in ANPC document 020-00074 "TLS Instructional Handbook: Maintenance Manual".

2.1 Equipment and Configuration

Performing the calibration and validation procedures requires setting up the Flight Test Monitor (FTM) and making the communication connection with the theodolite and its modem [Figure 2-1].

2.1.1 Flight Test Monitor (FTM) Equipment

The standard TLS Installation/Calibration kit [Figure 2-1] consists of the following equipment:

- Laser theodolite
- Tripod
- Two serial converters with Cat 5 connecting cable.
- Serial cable to connect the base station MIU to a serial converter.
- Serial cable to connect the theodolite to a serial converter
- Tape Reel
- Two hand-held short-range radios

Figure 2-1 FTM Serial Communications Link – actual serial converter may be different than shown

2.1.2 Pre-testing the FTM Assembly and Theodolite Data Link

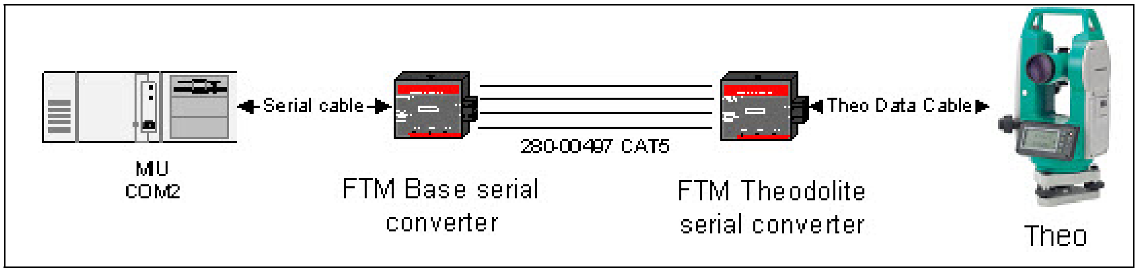


Figure 2-2 FTM Communications Link

Figure 2-3 FTM Communications Link serial converter – alternate part number to accomplish same purpose.

NOTE

Always start the FTM application from the MI, by selecting "Start FTM" in the MI Tools menu. Starting FTM from Windows Explorer will result in a failure for the MI and FTM to communicate correctly.

1. Place theodolite and other assembly components on a level table near the MIU [Figure 2-1]. Level the theodolite so that it will transmit test data, if it is not level it will not transmit data over the serial connection.
2. Use the serial cable with the DB-9 connector to attach the FTM base serial converter to the MIU serial port number two labeled "Com 2" [Figure 2-4]. For TTLS, this converter may already be installed

3. Locate the CAT5 ethernet cable spool and attach to the interface panel on the outside of the conex and the theodolite serial converter.

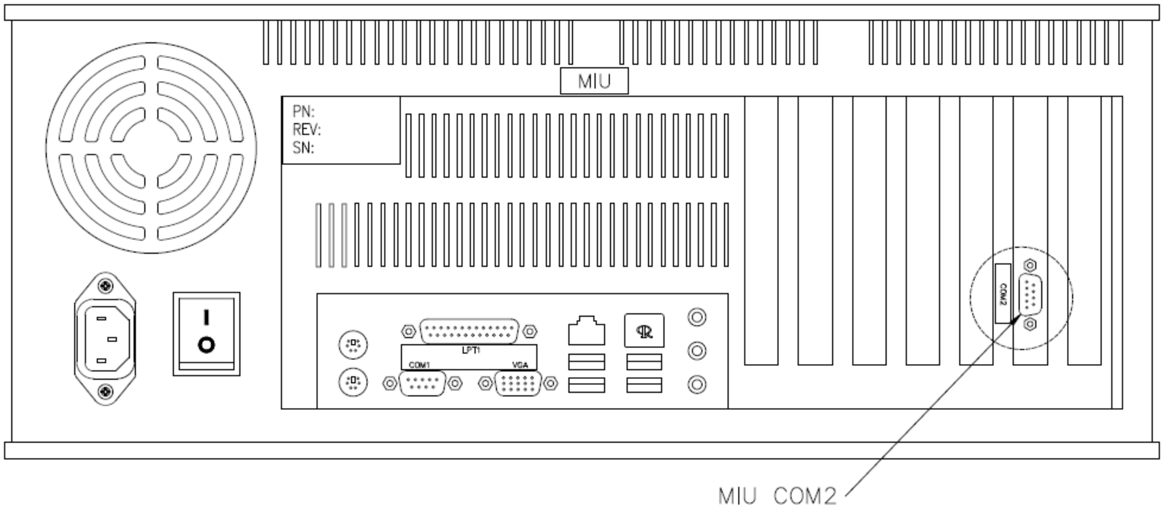


Figure 2-4 MIU Communications (COM) Port 2 DB9 connector – actual computer may be different

4. Use the serial data connector as shown in Figure 2-5 to attach the theodolite serial converter to the theodolite.

Figure 2-5 Theodolite Serial Data Connection

5. Power on the theodolite.

6. Unlock theodolite horizontal and vertical vernier knobs (see Figure 2-8) and rotate theodolite eyepiece through a full 360-degree rotation both horizontally and vertically. If a non-laser theodolite is used, only the vertical rotation is required.

7. Level the theodolite. The theodolite will display a message “out of range” and will not transmit data if it is not level.

8. On the MI, select Start FTM in the Tools menu.

9. Select "RS-232..." [Figure 2-9] from the FTM theodolite menu. In the dialog that opens, select COM2 and 9600 baud. Click "OK" to save and exit back to main FTM window.

10. Click the green "P" (Ping theodolite) [Figure 2-9] button and verify elevation and horizontal data is displayed in the FTM ping test window. Close FTM ping test window.

The FTM Link is now configured and the theodolite can be moved to the position next to the runway for TLS calibration and flight check procedure.

2.1.3 Theodolite Setup for Calibration or Flight Inspection

Use this setup procedure for both the calibration and the flight check procedures. Before you begin the procedure, contact Air Traffic Control (ATC) to coordinate a time when it is safe to work next to the runway.

1. Position the theodolite at the pre-survey location along the runway such that the theodolite eye piece will be located on the glide slope and centerline of the approach. This location can be on runway centerline if permitted by ATC and airport operations (low traffic time of the day). If the location is offset from centerline, an offset theodolite approach must be configured in the approach.dat file.
2. Center the tripod over the designated position with one leg pointing towards the approach direction.

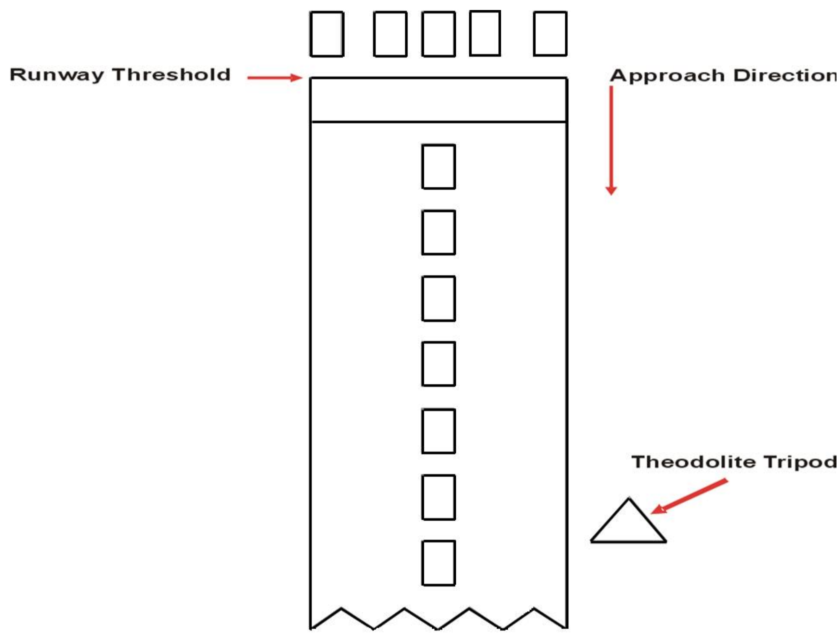


Figure 2-6 Runway Theodolite Positioning Example

3. Attach theodolite to the tripod using the thumb screw on base of theodolite.
4. Setup the theodolite over the survey mark. See 020-00073 “FTM theodolite position survey” if the survey mark has not yet been measured. One tripod leg should be pointed toward the approach direction so that the operator can stand behind the theodolite to track the aircraft through the eyepiece. Make the height of the theodolite eyepiece the same height as the eye height of the person who will operate the theodolite. Rotate all three theodolite adjustment wheels counterclockwise to the stop point.

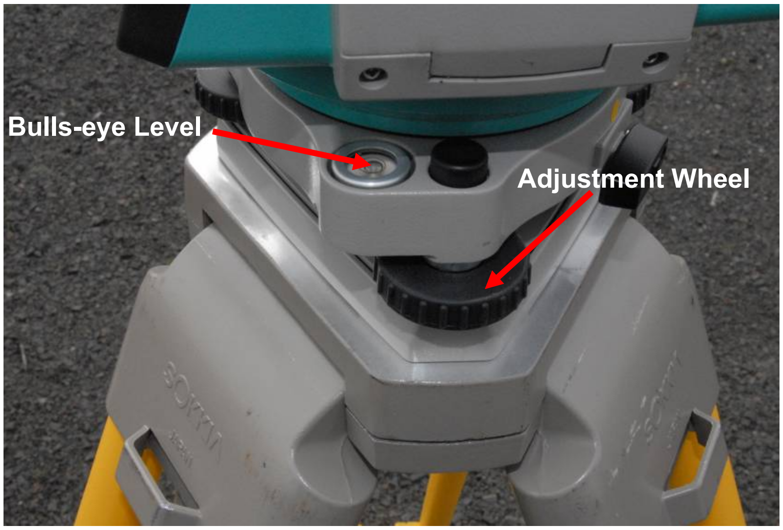


Figure 2-7 Theodolite Bulls-eye Level and Adjustment Wheels

5. Level the theodolite using the tripod and the bulls-eye bubble as the leveling reference [Figure 2-7]. Ensure the tripod legs are firmly planted in the ground. Use the leg length adjustment for course leveling and then the wheel adjustment only after the bubble level is close to center.
6. Use the adjustment wheel of the theodolite's left base to level the instrument. [Figure 2-8]. Use the Fine Alignment Bubble as the leveling reference until level is close to center
7. Use the other leveling wheels to fine tune the level of the theodolite until it is centered as indicated by the Fine Alignment Bubble [Figure 2-8].

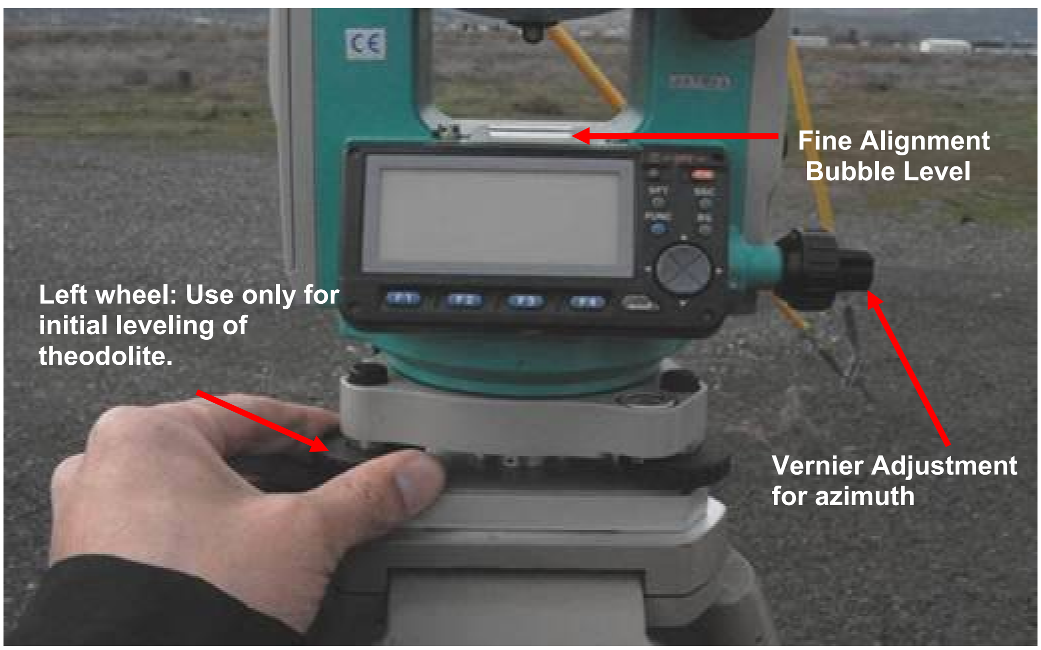


Figure 2-8 Fine Adjustment Leveling

8. Perform steps 2 through 8 of section 2.1.2 to connect the serial communications link between the TLS MIU and the theodolite.
9. Power on the theodolite and rotate the eyepiece 360º on the vertical axis (and horizontal if using the laser theodolite) to initialize modem communication.

10. Zero the theodolite azimuth on the previously surveyed backsight. See 020-00073 "FTM theodolite position survey" if the backsight has not yet been measured
11. Ensure enough vertical vernier adjustment to reach at least 7 degrees up, and that horizontal vernier range of adjustment is approximately centered on the surveyed backsight.
12. Using the hand-held radio, notify the TLS base station technician that the theodolite is ready.

2.1.4 FTM Setup

1. On the MI, exit FTM if it is running.
2. On the MI, select "Start FTM" from the Tools pull down menu.
3. The Flight Track Monitor (FTM) window will display [Figure 2-9].

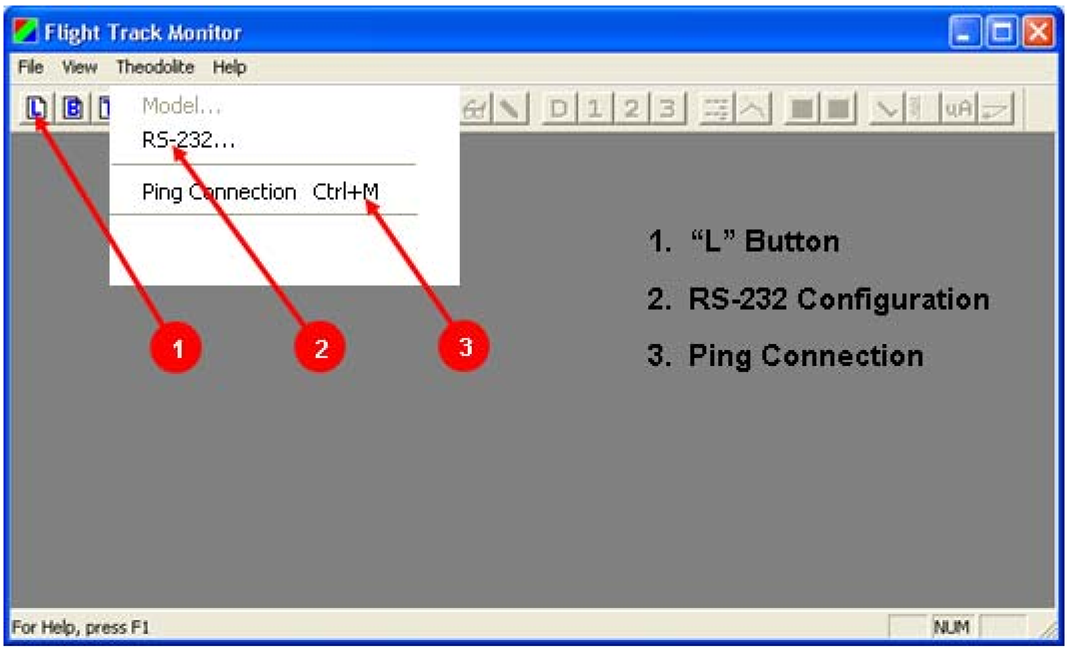


Figure 2-9 FTM Window

4. Select "RS-232" from the FTM window's theodolite pull down menu
5. Select the glide slope location area and input the distance from runway threshold to the theodolite position [Figure 2-10]. Note that the sign should be negative; record to the nearest tenth of a foot. For example, in the Figure below the number is -636.4.

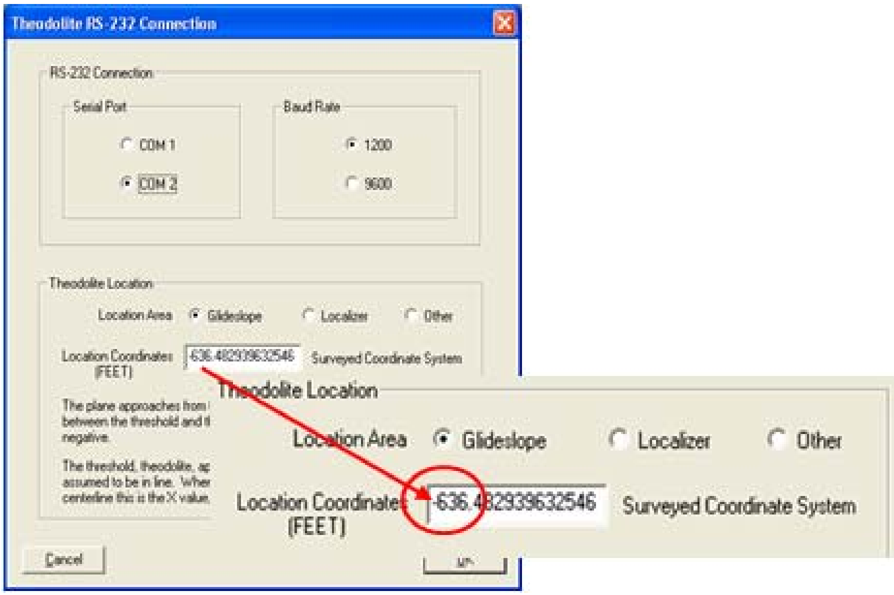


Figure 2-10 RS-232 Configuration Window

6. Click on the OK button to close the RS-232 setup window.
7. Verify with the theodolite operator that the theodolite assembly is completed.
8. Select Ping Connection from the theodolite menu control button (see Figure 2-9).
9. Verify Ping is shown on the FTM display and therefore correctly communicating.
10. From the FTM window select the "L" button.
11. Optionally, enter preflight information in the log entry form for weather, aircraft, pilot, type and purpose of flights. Click OK. This opens a new FTM document.
12. The FTM is ready to continue with the calibration procedure below. Close the FTM window.

2.2 Calibration Procedure

2.2.1 Calibration Profiles

The procedure for calibration includes system configuration, pre-flight planning, and calibration profiles.

2.2.1.1 Base Station Configuration

1. On "Settings" in the MI make the following settings:

Table 2-1 Base Station Configuration: MI Settings

Data Recording:	Enable Raw, RAM-Track, GTU files, - RAM Files BIT is not checked
First Character of Raw File Name:	Enter the first letter of your location
Azimuth Volume - Edges:	40 degrees
Elevation Volume - Lower Edge:	0.6 degrees
Elevation Volume - Upper Edge:	7.0 degrees
Localizer Range:	20 NM

2. Restart the system to effect changes.
3. Coordinate with airport authorities so that the theodolite operator can be next to the runway during the calibration flights.
4. Ensure that sections 2.1.3 Theodolite Setup and 2.1.4 FTM Setup have been completed.

2.2.1.2 Pre-Flight Planning

Brief the TLS test operators and the pilot(s) on the following information:

1. Five data collection profiles are required for the calibration data set:
   a. Two at constant altitude (level crossing) on the approach centerline at the greater of 1000’ AGL or lowest safe altitude
   b. Two at constant altitude (level crossing) on the approach centerline at the glide slope intercept elevation (if known) or 1600’ AGL
   c. One localizer arc ranging from 35 degrees to one side of centerline to 35 degrees to the other side of centerline, flown at 5 NM from threshold and constant altitude of 1600' AGL (or 500' above minimum safe VFR altitude)
   d. One localizer arc from 35 degrees to one side of centerline to 35 degrees to the other side of centerline, flown at 10 NM from threshold and constant altitude of 2000' AGL (or 500' above minimum safe VFR altitude)
   e. One profile from 8 NM, starting at altitude 1600', near the glide slope intercept (approx. 5 NM from threshold for a 3.0° approach), the pilot descends on glide path to threshold (this approach could be to full stop).

      *There will be no TLS guidance during this approach, so the pilot only needs to fly the approach to the best of their abilities. Maintaining a smooth descent is more important than maintaining glide path

   f. Low visibility special instructions

   If the visibility conditions are expected to be poor and prohibit being able to see the aircraft at a distance of 8 to 10 nm, then the low visibility procedure should be used as follows. The procedure is more interactive in that the helicopter climb through the service volume must be timed in coordination with the theodolite operator confirmation that visual contact has been established.

      1. A helicopter should be used to begin the profile on centerline at a distance of 4nm and elevation of 200 feet height above ground. The height of the helicopter will therefore be close to 0.5 degrees relative to the glide path point of intercept with the runway in order to start the calibration data below the lowest edge of the glide slope service volume.
      2. After the theodolite operator confirms that the helicopter is visual and the theodolite operator is tracking, then the TLS operator will ask over the airband radio for the pilot to climb to an elevation of 1,200 feet while maintaining course near runway centerline.
      3. Repeat this procedure twice, verify good TLS data was saved and then calibrate TLS using the SysCal software tool.
      4. If supported by the atmospheric visibility conditions at the time of the flight, ask the pilot to conduct a normal approach from 6nm intercepting the glide slope and following the glide slope and localizer to conduct a low approach to 100 feet while the theodolite operator tracks the aircraft.

2. All profiles (except for localizer crossings) need to be near the approach centerline and heading, and each at its respective constant altitudes (with the exception of the last approach). If the

theodolite location at the edge of the runway is to be used for calibration, the pilot(s) will be asked to fly a course to that edge of the runway - not to runway centerline.

3. The course shall be maintained until the TLS operator requests the pilot to break off the approach and set up for the next profile.
4. Each calibration profile will be considered complete when the theodolite operator has successfully tracked the aircraft, and base operator has recorded the FTM file from 1 degrees elevation to at least 6 degrees.
5. For any profile, verify the quality of collected data before proceeding. Review the items in the following verification list:
    - Did the pilot adequately fly the desired profile?
    - Did the theodolite operator successfully track the aircraft?
    - Was the .dif file saved, and does it have measurements?
    - Was the .raw file saved, and does it have measurements?
6. Create notes using Microsoft Notepad or its equivalent and save them along with the data.
7. Radio communications between the base station and the calibration aircraft shall be maintained at a radio frequency to be agreed to prior to the profiles.
8. The pilot is responsible for all safety decisions during the calibration profiles. If an approach has to be aborted due to safety reasons, the pilot will repeat the approach.
9. The pilot will request and receive an IFR transponder ID code from ATC and relay this code to the TLS operator.
10. Coordinate radio communications between the FTM operator and the TLS operator on a ground-to-ground radio frequency that will not interfere with ground-to-air communications.
11. The calibration aircraft must be equipped with a transponder, ILS receiver, air-to-ground radio, and landing lights. The pilot will turn on the landing lights during each approach to facilitate tracking with the theodolite.
12. The pilot shall be briefed to ignore any needle indications while the TLS is operating prior to calibration process being completed.

WARNING

The TLS operator(s) brief the pilot(s) about general concepts of TLS operation, and review how the acquisition - track aspects of the system work. If the pilot receives needle indications during calibration, the needles should be ignored.

2.2.1.2.1 Calibration Profiles

1. Coordinate the time of aircraft flight so that no aircraft air time is lost while waiting for the FTM-Base communication links to be established and tested.
2. If not already done, then in the FTM window, click on the "L" button to open a new FTM document and enter the preflight information. This preflight information can include specifics about the weather, aircraft, pilot, type and purpose of flight, etc. Click on "OK."
3. Instruct aircraft to make a procedure turn onto the first approach (level at minimum altitude) at between 8 and 10 NM, and to inform the TLS operator when the aircraft is starting the level run.

4. From the MI Tools menu, select "Test Mode" [Figure 2-11]. Switch the front panel power switch of the GTU(s) to off.

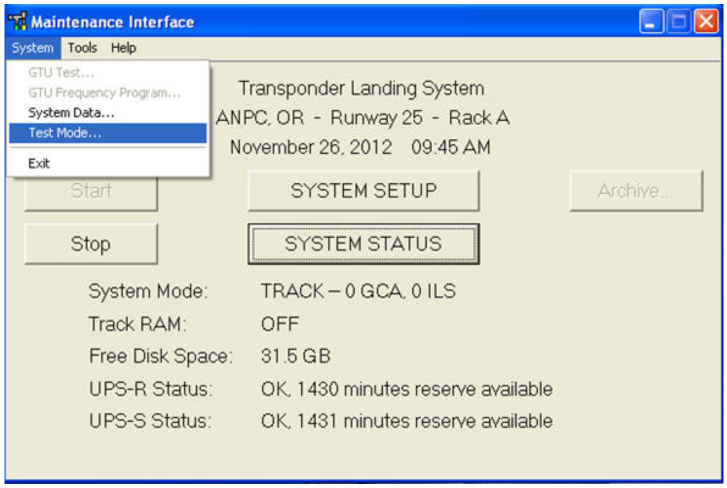
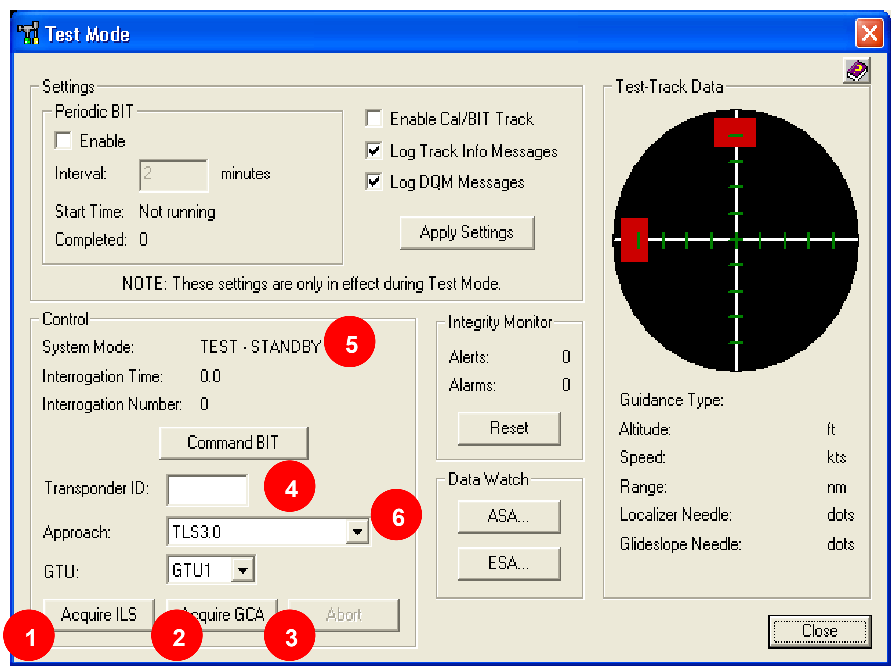


Figure 2-11 Test Mode Drop Down Menu

5. In the Test Mode window, select the Offset Approach and Transponder ID that will be utilized for the calibration profiles. The Offset approach for calibration is the offset from runway centerline to the edge of the runway [Figure 2-12].



1. Acquire ILS Button
2. Acquire GCA Button
3. Abort Button
4. Transponder ID
5. System Mode
6. Approach

Figure 2-12 Test Mode Drop Down Menu

6. On the MI in Test Mode, after the pilot indicates that the approach has started, with aircraft approximately lined up on the calibration approach, heading inbound at the first altitude, and theodolite operator searching for aircraft, click the "Acquire" button [Figure 2-12].

7. Verify that the TLS acquires a track on the aircraft. The Test Mode window will display "TRACK - LOC ONLY" beside the System Mode field.

8. When the theodolite operator reports over the radio that he is visually tracking the aircraft with the theodolite, the "Start Record" button on the FTM pop up dialog [Figure 2-13]. Note that the “start record” window may be anywhere on the desktop, depending on where the last user pushed the window.

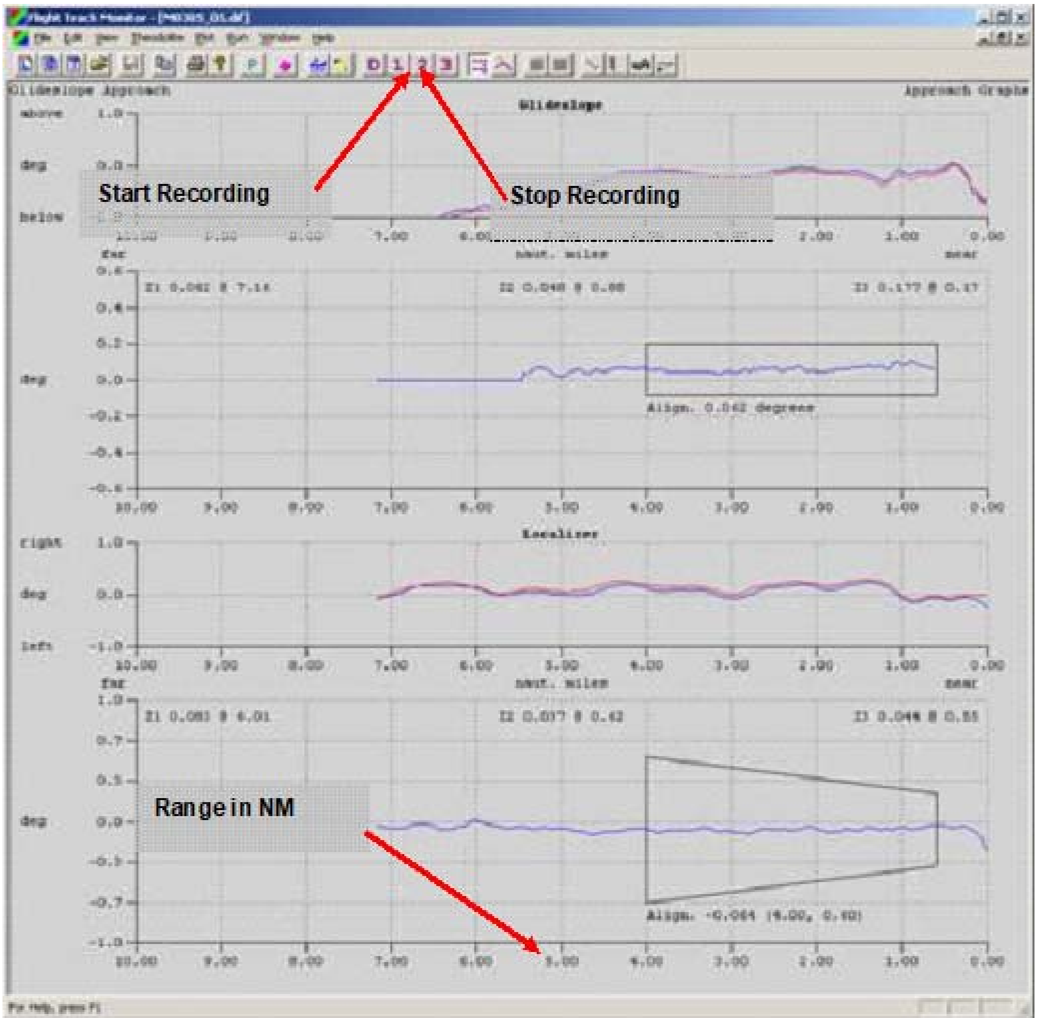


Figure 2-13 FTM Record Control

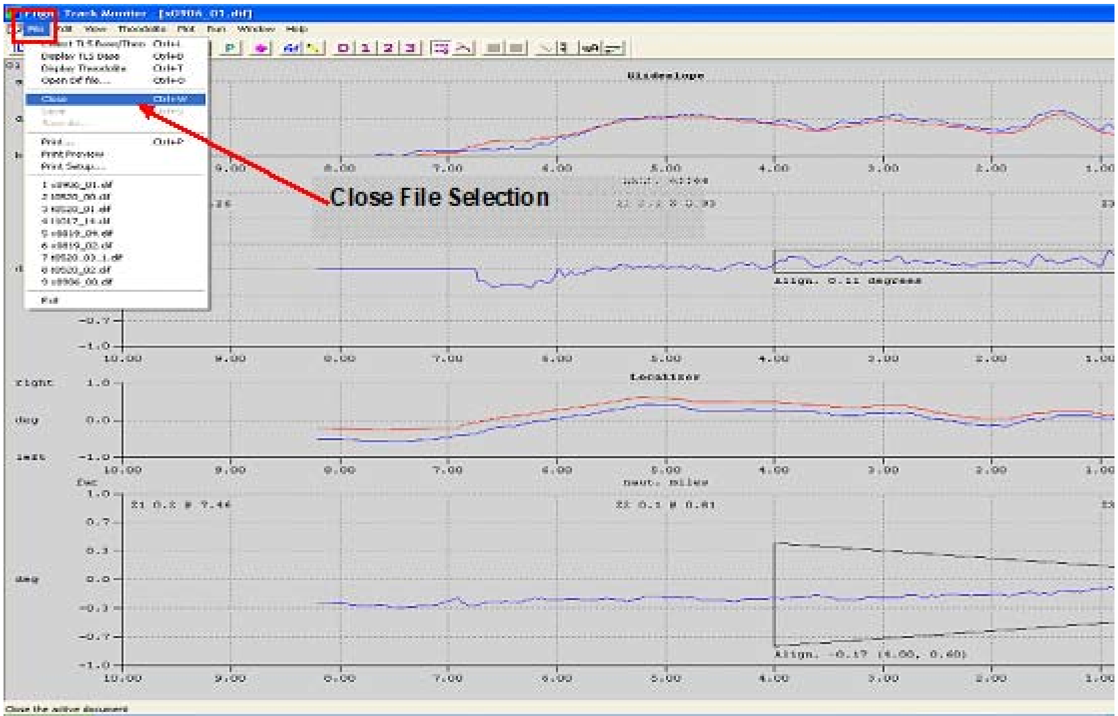


Figure 2-14 FTM Close File

9. If the theodolite operator has not achieved visual track prior to the aircraft reaching about 1 degree elevation or 6 NM abort the flight and restart the procedure.
10. Let the FTM system track the aircraft until the following occurs:
    a. The theodolite operator reports loss of visual track or
    b. The red line on FTM glide slope track passes the upper edge of the display window or
    c. The tracks on the FTM window have passed range of 0.5 NM or
    d. The aircraft aborts the procedure.

    The purpose of this procedure is to have FTM recording while the theodolite is actively tracking the aircraft.
11. Click on the "Stop Recording" button in the FTM window [Figure 2-13].
12. Log the file name of the file that was just recorded. Do not change the default filename. The calibration software in the steps below uses the default file name convention.
13. In the Test Mode window, click on the "Abort" button [Figure 2-12].
14. Inform the pilot that the profile has been completed and to continue on to the next profile.
15. Inform the theodolite operator that the flight has been discontinued and to re-center equipment.
16. In the FTM window, click on the "File" menu button to save the file [Figure 2-17]:
    a. Choose the "Close" option to close the recording file
    b. Make sure the file is saved to the directory \\base2\c\run\raw
    c. Click on "Yes" to save the recorded file. Do not change the default filename. The calibration software in the following steps uses the default filename convention.
17. Exit the Test mode window by clicking on "Exit Test mode" button.

18. Repeat steps 3 through 17 for the other profiles. If there are problems with any profile, repeat that profile so that the proper number of successful data sets is recorded. Add detailed annotation to the notes section of the data sheet.
19. Record validation that profiles were completed, and the raw file numbers for the successful calibration profiles.
20. Click "Close" on the MI Test Mode window, and wait for system to display "Track" in the status window.
21. On the MI, stop the system software.

2.2.2 Calibration File Generation

The objective of this step is to generate the calibration file that provides precise localizer and glide slope alignment.

1. On the MIU computer, start the SysCal tool. It is located in the "C:\Tools" directory and is called SysCal.exe. Double click on the object to start it up. See the tool's help menu for additional instructions.

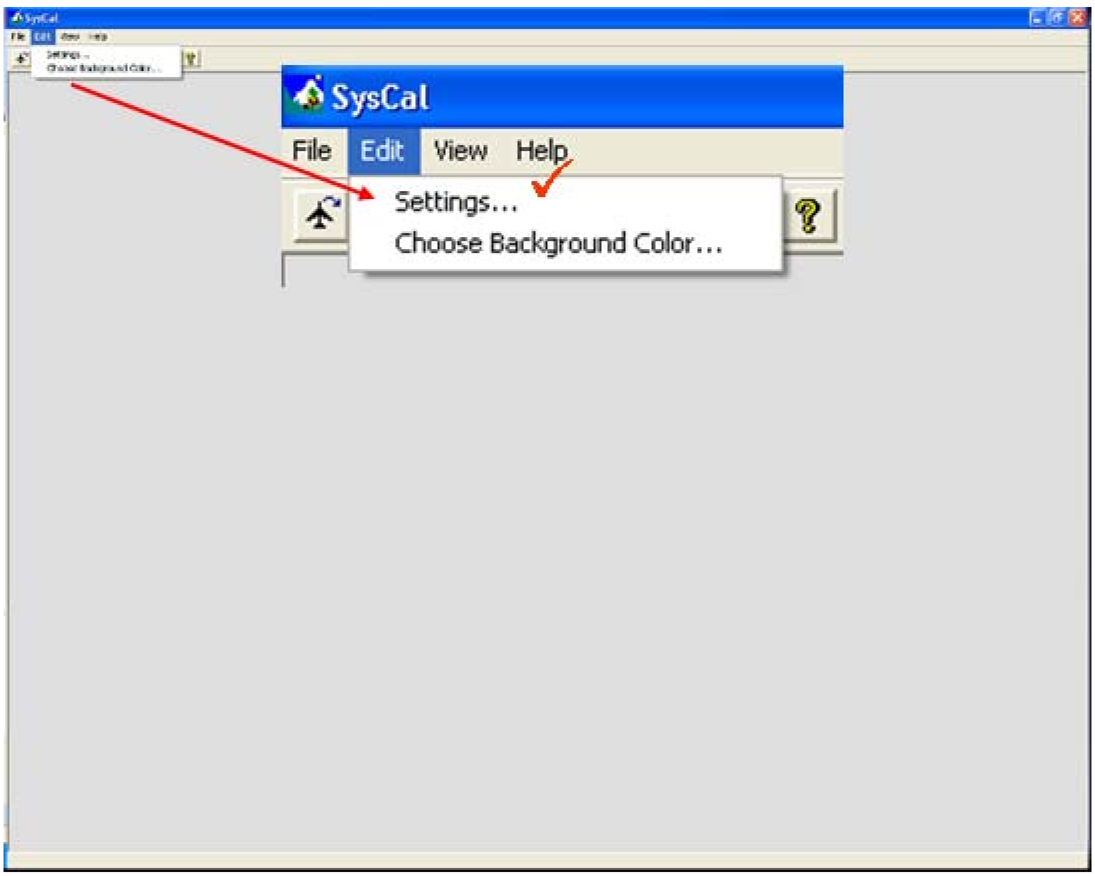
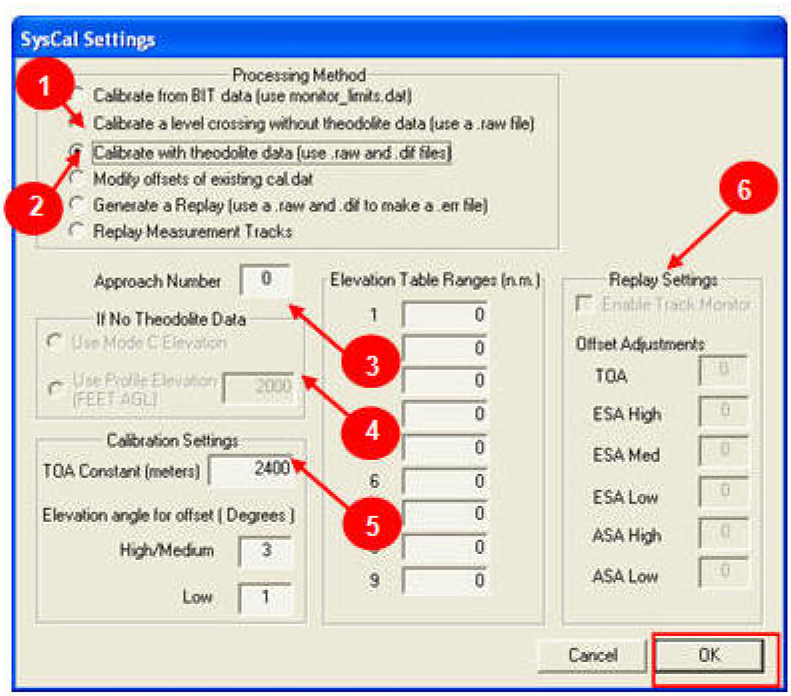


Figure 2-15 SysCal Settings

2. Under the "Edit" menu, open the "Settings" dialog. Select the "Calibrate with theodolite data" processing option [Figure 2-16, number 2].



1. Calibrate using level crossing
2. Calibrate with theodolite data
3. Approach number
4. Level crossing height
5. TOA constant
6. Replay settings

Figure 2-16 SysCal Settings Dialog

3. Enter the selected approach number. The number is equal to the row of the approach from the approach selection box minus one.
4. Enter the calculated “TOA Constant". This is the calculated system delay for the TOA measurement and is used to calibrate the range. It is typically around 2400 if the cable runs out to the ESA and ASA are fairly equivalent.
5. Click on “OK” to save and close the Settings dialog [Figure 2-16].
6. Click the airplane button [Figure 2-17]. A file selection box will be displayed. The files (.raw) used to calibrate are normally selected from the "\\Base2\c\Run\Raw" directory. Using the flight log to decide, select the files that captured data for the level glide slope crossings and the straight-in approach files. You can select multiple files using the shift and control keys. These files will be used to calibrate the system.

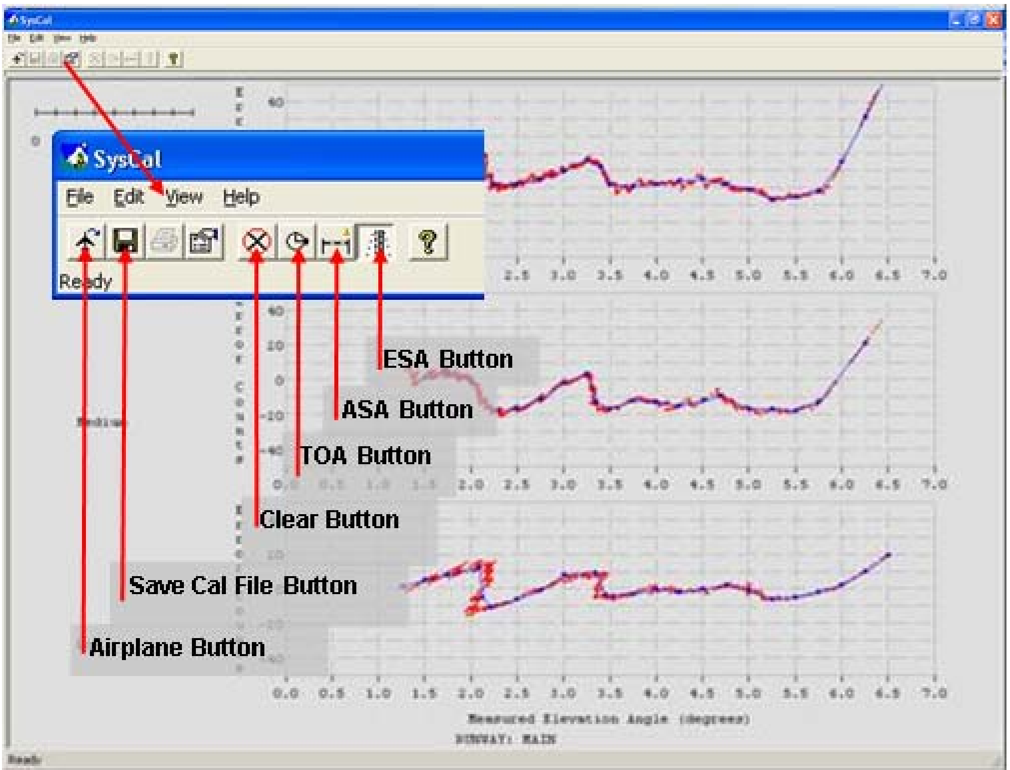


Figure 2-17 SysCal Tool

When the progress bar is done the calibration data for the file is initialized based on the data shown. Data for the different types of measurements from the different sensor types can be represented. In general, the data presents a difference plot between the system's calculated position and the theodolite or truth position for the aircraft.

7. Start by clicking on the TOA button [Figure 2-17]. The green data plot should trend horizontally until approximately 3 NM. The blue line running through the green data represents the mean of the data between roughly 6 and 8 NM. This data is typically noisy but overall horizontally level. Erratic, non-horizontal data could indicate a system configuration issue. Validate that the data trends horizontal.
8. Next, click on the ASA button [Figure 2-17]. On the upper two plots the red data should also be primarily horizontal. Again the blue line represents the mean of the data between 6 and 8 NM. Look for major inconsistencies or continuous drifts in the data. Typically this data is consistently horizontal with some minor fluctuations or oscillations. Typically there is nothing to do here but validate that the data is horizontal in nature. The lower two plots are informational only.
9. The critical vertical data comes from the ESA sensor's phase measurements. Click on the ESA button [Figure 2-17] to display the calibration plots pertaining to glide slope. See Section 3.9.1 of the ANPC TLS Operators Manual 020-00072 for a close-up view of the calibration curve and dotted line.
   - The red dots display the error data for the high, medium, and low phase measurements.
   - The blue line with dots displays the best fit for the data and is called the calibration curve. Each dot on this line is a value that gets stored in the calibration file. Drag the dots to improve data fit.

Make any necessary adjustments in the ESA plot by left clicking and holding down on the blue curve points and moving the point to the red plot. Delete unnecessary points by pressing shift + left click on the point. Right click where you want a point inserted to add a point. The goal is to represent the red data points with the blue curve. Ignore obvious outlying data points or points that are on a secondary angle-of-arrival cycle.

Typically, the tool generates a calculated curve. This curve can be used to examine the performance of the current flight data through replay runs. A replay will use the system software along with the new calibration data to rerun the tracking solution for a flight with an aircraft under guidance.

Do not change the default filename because the calibration software in the steps below uses the default file name convention.

10. Save the new calibration file by using either the 'save', or 'save as…' command under the file menu. Do not change the default filename because the calibration software in the steps below uses the default file name convention.

11. A combination of the SysCal and Look tools will be used to verify that the calibration successfully aligned the measurements:

    - The Look tool will reveal any calibration deficiencies between the different measurement tracks.
    - The replay function of SysCal will be used to adjust the alignment of the tracks and to quantify the resulting performance of the guidance track.

12. In SysCal, from the Edit menu bar option, select Settings [Figure 2-16]

13. On the Settings dialog, under the Processing Method section, select the "Generate a Replay" radio button [Figure 2-19].

14. Ensure all the numbers under the Replay Settings section get zeroed out and ensure that the "Enable Track Monitor" check box is deselected [Figure 2-16].

15. Click on OK to save the new selections

16. Use the Airplane button [Figure 2-17] to reselect the file that generates the calibration data with the longest flight time and range. Generally, this is the straight-in approach. The replay function will automatically begin.

    Use this step of the process first to ensure that the system alignment performance is adequate for both glide slope and localizer. On the left hand side of the plot is a set of numbers for the alignment and structure of the elevation and the azimuth. Note these numbers. The alignment error for the glide path angle must be less than .05 degrees. The alignment error for the localizer must be less than 0.1 degrees.

17. To adjust the alignment for the azimuth or ASA select the "Settings" option from the Edit menu item on the SysCal tool main page.

18. Enter the numbers noted in step 16 into the ASA High and the ASA Low data fields [Figure 2-18]. The numbers have a relationship of 4:1 for high to low. So, if you put 40 in the high, you would put 10 in the low if you want the adjustments to track between measurement types. A negative number will bring the blue line up. A positive number will push it down. The goal is to have the blue and redlines overlap as nearly as possible.

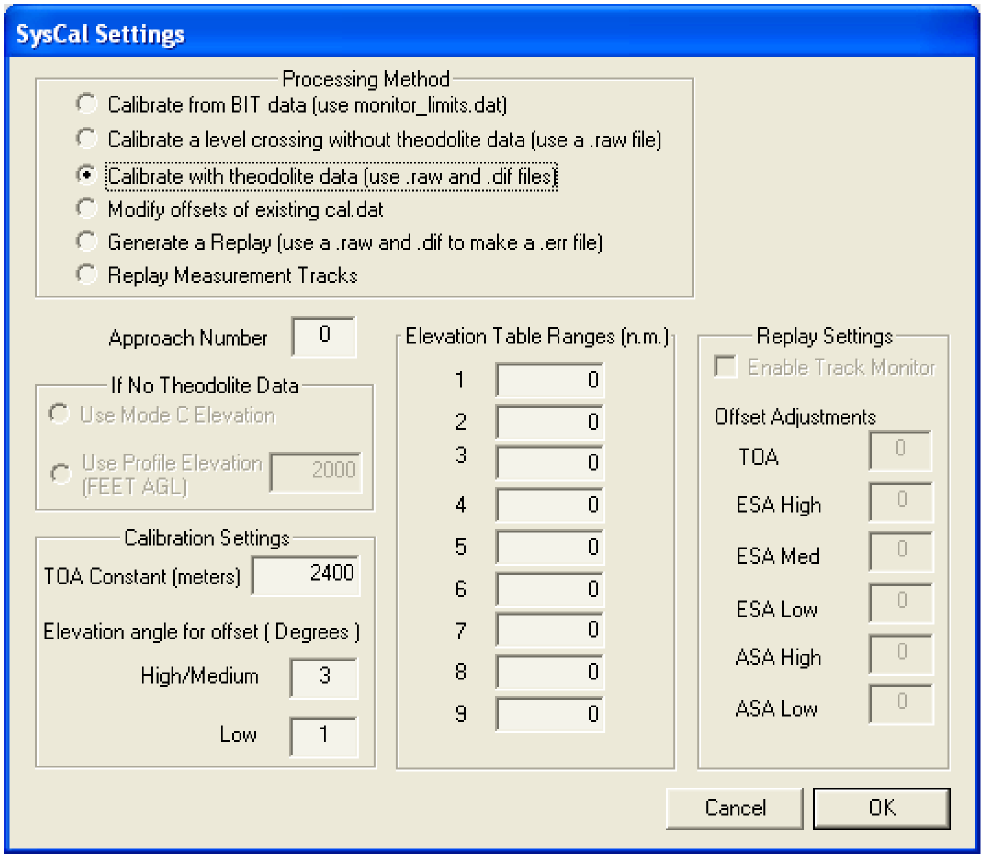


Figure 2-18 Settings Dialog Screen

19. Enter the adjustments, and then click OK to save. Use the airplane icon to select the same file that was processed the last time. Visually inspect the data lines to see how they overlap and alignment values to see if they are improved. Repeat steps 18 and 19 until the alignment is as close as possible.

20. Align the glide slope after azimuth alignment. Enter the numbers into the ESA High, Med and Low data fields under the Replay Settings section of the Settings dialog [Figure 2-19]. Again, the numbers have a relationship of 4:2:1 for high to med to low. Which means if you enter a 40 in the high field then a 20 goes in the Med field and a 10 goes in the Low field if you want similar adjustments to each measurement types. A negative number will bring the blue line up. A positive number will push it down. The goal is to have the blue and redlines overlap as nearly as possible.

21. After entering the adjustment click on OK to save. Using the airplane icon select the same file that was processed the last time. Visually inspect the data lines to see how they overlap and alignment values to see if they are better. Continue steps 19 and 20 until the alignment is as minimal as possible.

22. Save the calibration file. Do this selecting the Save button from the main SysCal tool dialog [Figure 6-35]. After this point, whenever a replay is run, it is important to be sure that on the Settings dialog, all of the fields under the Replay Settings section have 0’s in them [Figure 6-36]. Otherwise the alignment will move for the next replay run. One way to be sure the fields are reset is to use the Clear button on the main SysCal dialog [Figure 2-16].

23. Start the Look tool on the MIU computer. Located in the “C:\Tools” directory, it is called Look.exe. Double click on its icon object to start it up. See the tool's help menu for additional instructions. Details for reviewing measurement tracks with the Look tool are presented in section 2.2.3 .

24. Examine the calibrated measurement track in Look and verify that their agreement is within ½ degree.

25. If more alignment adjustment is necessary, repeat steps 18-19 for localizer and 20-21 for glide slope until you are satisfied. Be aware that the overall path alignment is determined by the high measurement track. If necessary, just make adjustments to the medium and low to bring them in agreement with the high.

26. Save the new cal.dat by clicking on the save button on the menu bar [Figure 2-16]. This saves the new "cal.dat" file over the existing file on "Base2" in the "c:\run" directory. Exit the SysCal tool.

27. On the MI, open the SendToBases tool. It is located in the “C:\Tools” directory and is called sendtobases.exe. Double click on the object to start it up.

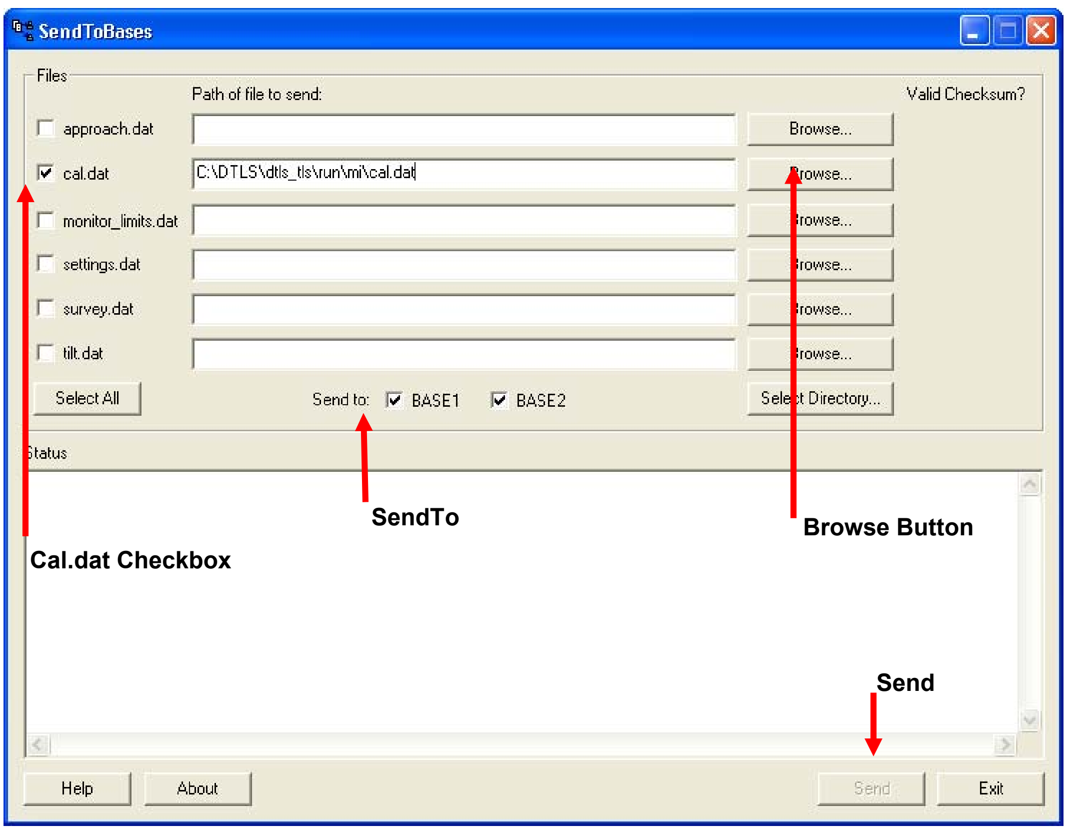


Figure 2-19 SendToBases Dialog Screen

28. Select the check box for ONLY the "cal.dat" under the "Files" section. Click on the corresponding "Browse..." button [Figure 2-19].

29. On the Browse Dialog, from the "Look in:" pull-down menu, select the "\\Base2\c\run" directory. From this directory click to highlight "cal.dat" followed by clicking the "Open" button.

30. On the SendToBases Dialog, select the BASE1 checkbox and uncheck BASE2 checkbox. Click the "Send" button at the bottom of the "SendToBases" tool, then click "Exit" to exit the tool after the "Status" window shows that the process has completed.

31. Record the validation of successful calibration file generation in the facility log book.

2.2.3 Calibration Validation Verification (Look Tool)

Use the Look tool as a quality control to verify that the calibration procedure has successfully aligned the calibration measurements.

1. On the MIU computer start the Look tool (Look.exe) located in the “C:\Tools” directory. Double click on the object to start it up. See the tool's help menu for additional instructions.
2. Use the tool to open the raw file on Base2 that was used for calibration.
3. Select measurement track buttons for the low, medium and high resolution measurements from the Track Processing Toolbar as shown in Figure 2-20.
4. Select the Calibration Effect Toggle Button [Figure 2-21] that allows viewing the effect of calibration. Verify that the measurements are within the following tolerances of each other:
   - The high and medium are within ± 0.75 degrees.
   - The low is within 1.0 degree of both the high and medium.

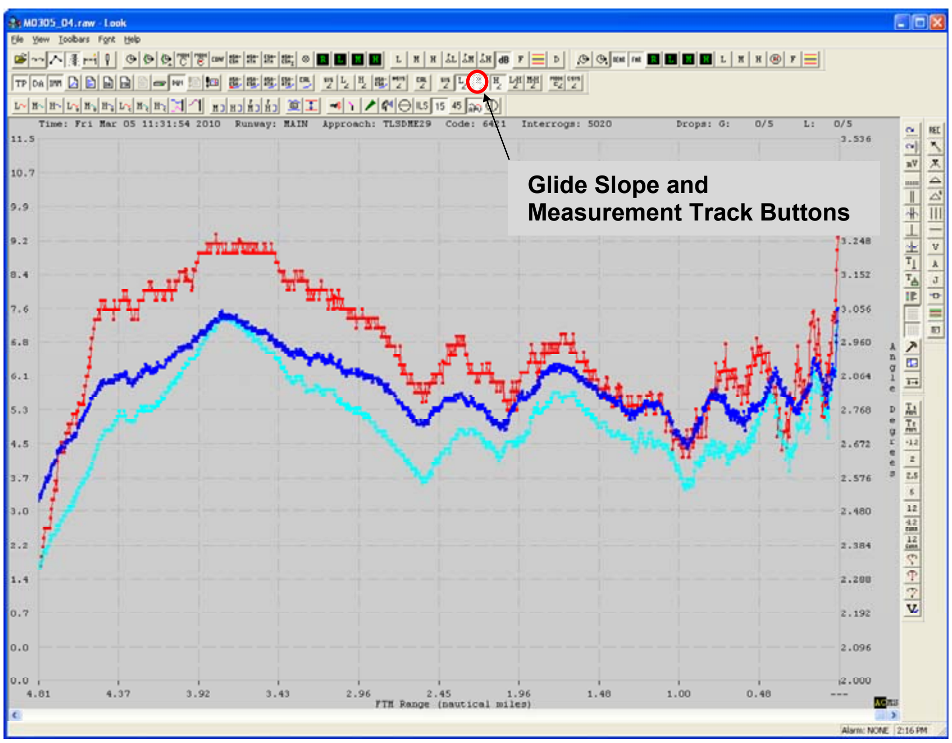


Figure 2-20 Look Tool: Measurements Tracks

5. If any of the measurement tracks need adjustment, use the SysCal tool to modify cal.dat and then repeat steps 2-5.

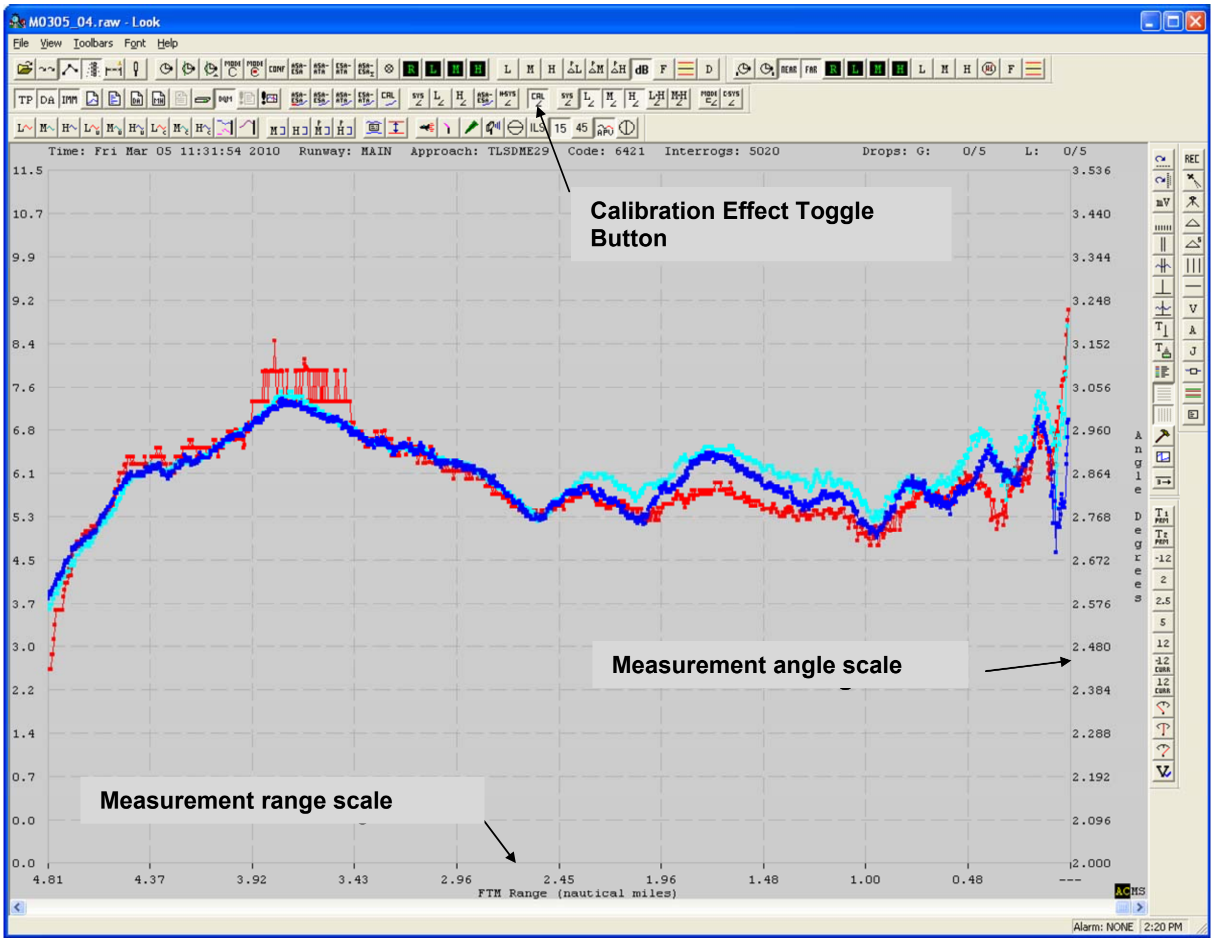


Figure 2-21 Look Tool: Calibration Applied to Measurement Tracks

2.3 Alternative Calibration Procedure – no theodolite reference

This calibration method uses a single aircraft flight profile and does not require a technician to visually track the aircraft with the theodolite. This procedure should be used when hazy conditions prevent using a theodolite.

2.3.1 Calibration Quality: Course/Altitude

The quality of this calibration process relies on the pilot to fly and maintain a specific course and altitude for the entire profile. Instead of calibrating the system to an independent truth source (the theodolite), the system will estimate the position of the aircraft based on the expectation that the pilot will fly straight.

This procedure may be used as an alternative to the above procedure with theodolite if theodolite operations cannot be performed.

2.3.2 Adjustment/Verification

Final adjustment and independent verification of the localizer and glide slope parameters must be accomplished by Flight Inspection prior to commissioning and operational use. Since no theodolite is used for the initial alignment verification, it is likely that more adjustment will be needed during flight inspection.

2.3.2.1 Base Station Configuration

On the MI, in "Settings," make the following settings:

Table 2-2 Base Station: MI Settings

Data Recording:	Enable Raw, Track RAM-Guidance, and GTU files
Azimuth Volume - Edges:	40 degrees
Elevation Volume - Lower Edge:	0.6 degrees
Elevation Volume - Upper Edge:	7.0 degrees
Localizer Range:	20 NM
Glide Slope Range:	0.0 NM

2.3.2.2 Pre-Flight Planning

The following information should be briefed to the TLS operators and the pilot:

1. One successful data collection profile is required for this calibration data set: a constant altitude profile on the approach centerline. Repeated flight profiles help to ensure calibration data. The lowest practical altitude above 1600' AGL and below 2000' AGL should be used. This flexibility in altitude is provided in the event the pilot has a preference due to airspace issues or the terrain prohibits an approach at lower altitude.
2. The pilot is responsible for all safety decisions during the calibration profiles. If an approach has to be aborted due to safety reasons, it will simply be repeated.
3. The pilot(s) should be cognizant that the localizer may not be aligned with the runway centerline and there will be no glide slope guidance received in the cockpit during calibration profiles.
4. This aircraft flight profile should be near the runway centerline and heading, and the required altitude, held steady by plus or minus 100 feet. For example, if 1600 feet is selected for the level run the pilot should keep the aircraft between 1500 and 1700 altimeter reading.
5. The course shall be maintained until the TLS operator requests the pilot to break off the approach and set up for the next profile.
6. This calibration profile will be considered complete when the base operator has recorded the raw file from at least 10 NM until 2 NM from the approach runway threshold.
7. For any profile, verify the quality of the data collected before releasing the aircraft. Consider these items:
    - Did the pilot adequately fly the desired profile?
    - Check the raw file to verify that the measurements are present using the look.exe software tool.
8. The calibration aircraft must be equipped with a transponder and air-to-ground radio.
9. Radio communications between the base station and calibration aircraft shall be maintained.
10. The pilot will request and receive an IFR transponder ID code from ATC and relay this code to the TLS operator.

2.3.2.3 Aircraft Calibration Profiles

1. Ask the pilot to establish the aircraft inbound, on the extended runway centerline at 12 NM from the runway threshold. The pilot should inform the TLS operator when the aircraft is starting the level run at the required altitude.
2. On the MI, select "Test Mode" in the "Tools" menu.
3. At the Test Mode window, select the centerline Approach and enter the Transponder code.
4. On the MI in Test Mode, after the pilot indicates that the approach has started, with aircraft approximately lined up on the calibration approach heading inbound at the required altitude, click the "Acquire ILS" button.
5. Verify that the TLS acquires a track on the aircraft within about 12 seconds. The Test Mode window will display "TRACK - LOC ONLY". (note that a pulse-rich code may be needed to ensure quick acquisition
6. When the aircraft's range to threshold reaches 2 NM, in the Test Mode window, click on the "Abort" button.
7. Verify that the data has been successfully recorded.
8. If there are problems with any profile, repeat that profile so that the proper number of successful data sets is recorded. Add detailed annotation to the notes section of the data sheet.
9. Record validation that profiles were completed, and the raw file numbers for the successful calibration profiles.
10. Click "Exit" on the MI Test Mode window, and wait for system to display "Track" in the status window.
11. On the MI, stop the system software.
12. Exit the MI.

2.3.3 Calibration File Generation

On the MIU, open the SysCal tool. See the tool's help menu for up to date instructions.

1. Perform the procedure from section 2.2.2 Note that an exception occurs during step 2: when you select the "Calibrate a level crossing without theodolite data" processing option. Be sure the correct approach number is selected. Enter the altitude the aircraft in the Crossing Elev field [Figure 2-16], in feet AGL. Click "OK".
2. On the MI, open the Look tool. In Look, open the raw file on Base2 that was used for calibration. Examine the alignment of the measurements and ensure that the calibration step has been adequately completed.
3. In the Look tool, select the calibrated measurement buttons for the low, medium and high resolution measurements. Then select the button that allows viewing the effect of calibration. Verify that the measurements are within ½ degree of each other. If a portion of a measurement curve differs from the others by more than ½ degree.
4. On the MI, in "Settings," return the glide slope volume setting to the default:

Table 2-3 Calibration File Generation: MI Settings

Data Recording:	Enable Raw, Track RAM Guidance, & GTU files
Azimuth Volume - Edges:	40 degrees
Elevation Volume - Lower Edge:	0.6 degrees
Elevation Volume – Upper Edge:	7.0 degrees
Localizer Range:	25 NM
Glide Slope Range:	12 NM

2.4 Settings Validation

1. On the MI, in "Settings", make the following settings:

Table 2-4 Settings Validation: MI Settings

Data Recording:	- Enable Raw and GTU files - Enable Track RAM Guidance - Enable BIT RAM
Azimuth Volume - Edges:	40 degrees
Elevation Volume - Lower Edge:	0.6 degrees
Elevation Volume - Upper Edge:	7.0 degrees
Localizer Range:	25 NM (or site localizer range)
Glide Slope Range:	12 NM (or site glide slope range)
Diagnostic BIT Schedule Interval:	10 seconds
Diagnostic BIT Schedule Retries:	3
Guidance Rate:	20 Hz
Surveillance Rate:	2 Hz

2. Record verification that settings have been updated.

2.5 Signal Performance Verification: Procedure and Criteria - Optional

This procedure can be used to prepare for formal flight inspection by a Civil Aviation Authority (CAA) or equivalent organization. The time and expense of conducting the procedure described below may not be justified if a CAA flight inspection has been scheduled and will take place. In that case this procedure is optional.

This section provides instruction and performance criteria associated with theodolite-based inspection of the uplink VHF and UHF transmission. Review ANPC document reference 300-000038 for commissioning and periodic inspection tolerances and limits.

The desired aircraft profiles are listed below with the Flight Inspection name for the profile e.g. ILS-#. Note that these profiles and the subsequent flight check need to be done above the mask angle.

- ILS-1: localizer crossing at constant altitude; the aircraft travels at constant altitude across the entire 70 degree width of the localizer service volume.
- ILS-2: glide slope crossing; the aircraft travels on the approach course centerline at a constant altitude.
- ILS-3: aircraft travels the approach on course and on glide path.

The truth reference used to measure TLS performance is the FTM. This system uses a theodolite, data link, and software installed on the MIU to record the TLS solution, optical track and TLS system error (difference between TLS and the theodolite). These FTM data are processed and used to validate the tolerances and limits in ANPC document 300-00038.

2.5.1 Localizer Flight Verification

2.5.1.1 Localizer Course Width and Symmetry & Clearance

The purpose of this check is to establish the course width sector and ratio between half-course sectors. The theodolite must be positioned at the apparent localizer location. While being tracked by the theodolite, the test aircraft flies an ILS-1, 70° arc across the localizer course at the lowest site-specific coverage altitude at a distance of 5 NM from the apparent localizer position. Data is recorded and processed by FTM to provide a computation of the course width, symmetry and 90/150 Hz clearances.

2.5.1.2 Localizer Alignment and Structure

The purpose of this check is to measure the alignment and structure of the localizer guidance signal. The alignment and structure checks are performed simultaneously. The theodolite must be positioned at the glide slope location and the position of the theodolite should be recorded as shown in figure 2-10. The localizer course is evaluated from the furthest point required by the approach procedure to the runway threshold. During an ILS-3, the procedural altitude is maintained until the glide slope is intersected. Data is recorded and processed by FTM to provide a computation of localizer structure in each zone and alignment.

2.5.2 Glide Slope Flight Verification

2.5.2.1 Glide Slope Path Width and Symmetry

The purpose of this check is to establish the path width sector, the ratio between half-course sectors and the angle at which .250 DDM of fly-up occurs below path. This check assumes a 0.7° (± 0.05) course width measured at the 75 µA points. The theodolite must be positioned at the glide slope location. The aircraft flies an ILS-2, level crossing on the approach course centerline at the 5NM glide slope intercept altitude i.e. 1600 AGL for 3 degree glide path. Data is recorded and processed by FTM to provide a computation of the course width, symmetry and 193.5 µA angle. The aircraft altitude of this check may be adjusted due to ATC request, weather, or to obtain the 193.5 µA angle.

2.5.2.2 Glide Slope Clearances

This check is performed to ensure that positive fly-up indications exist between the bottom of the path sector and obstructions.

- Clearance below path is checked by flying the localizer on course. It checks to ensure that adequate obstacle clearance exists and that 180 µA or more of fly up signal exists from the final approach fix to point C (or the missed approach point).
- Clearances above the path are checked to ensure that a positive fly-down indication is received above the path angle. Clearance above the path is checked by a level run, where the glide slope width and symmetry are measured. The TLS shall provide 150 µA or more of fly-down signal from the 4° intersection to point C (or the missed approach point).

2.5.2.3 Glide Slope Alignment and Structure

The purpose of this check is to measure the alignment and structure of the guidance signal. The angle and structure checks are performed simultaneously. This test requires a theodolite located at the apparent glide path point. The glide slope course is evaluated from the furthest point required by the approach procedure to the runway threshold. During an ILS-3 the procedural altitude is maintained until the glide slope is intersected. The pilot sends radio marks to the Base Station at the outer marker (or 4 NM from threshold if the outer marker does not exist) and at the runway threshold. Data is recorded and processed by FTM to provide a computation of the structure in each zone and the alignment in Zone 2.

2.5.3 Verification Procedure

1. The verification profiles will consist of seven successful data sets collected with an aircraft flying the following profiles. These profiles will verify the approach procedure and measure localizer and glide slope alignment, structure, width, symmetry and clearance.

   Three profiles with the theodolite set-up at the localizer position:

   - Two localizer crossings to measure localizer <u>Course Width and Symmetry & Clearance</u>. Fly one arc from left-to-right and the other from right-to-left. Fly these arcs at GS intercept range and altitude;
   - One standard approach flying the full procedure to measure <u>Localizer Alignment and Structure</u>. Fly this approach so that localizer is centered prior to GS intercept.

   Three profiles with the theodolite set-up at the glide slope position:

   - One glide slope crossing to measure <u>Glide Slope Path Width and Symmetry</u> Fly this approach at GS intercept constant altitude, centered on the localizer from 5 NM outside the GS intercept range until runway threshold (or until system operator indicates that approach is complete);
   - Two standard approaches flying the full procedure to measure <u>Glide Slope Alignment and Structure</u>. Fly these approaches so that localizer is centered prior to GS intercept.
     - ✓ In limited visual conditions, night operations may be necessary to get the best range possible.
     - ✓ Confirm that the theodolite offset is entered correctly in FTM settings.
2. Coordinate the time of aircraft flight so that no aircraft air time is lost while waiting for the FTM- Base communication links to be established and tested.
3. These profiles should be performed from the RCU with the TLS in "Track".
4. Brief the pilot and other test personnel on the profiles and procedures.
5. The pilot will request and receive an IFR transponder ID code from ATC and relay this code to the TLS operator.
6. Instruct aircraft to make a procedure turn onto the approach at the intercept altitude at between 8 and 10 NM, and to inform the TLS operator when the aircraft is starting the profile.
7. In the FTM window, click on the "L" button to open a new FTM document and enter the profile description. This profile description can include specifics about the weather, aircraft, pilot, type and purpose of flight, etc. Click on "OK".
8. At the RCU window, enter the Transponder ID that will be utilized for the verification profiles.

9. At the RCU, after the pilot indicates that the approach has started, with aircraft approximately lined up on the approach heading inbound at the intercept altitude and theodolite operator searching for aircraft, click the "Acquire" button.
10. Once the theodolite operator is visually tracking aircraft with the theodolite, click "Record" button on the FTM pop up dialog.
11. Wait until the aircraft reaches threshold or leaves the guidance volume, depending on the profile, at which point the system will automatically transition to diagnostic.
12. In the FTM window, click on the "X" button to close the plot document and click yes to save the dif plot.
13. Record that the profile was completed, the raw file number, and any notes that can be applicable to the understanding of the collected data (Flight type, any unusual aspects of the profile, FTM operator errors if reported, etc.).
14. Repeat steps 6-13 for the other verification profiles.
15. On the MI, stop the system software.
16. Perform the data archive procedure (see document #020-00074 TLS Maintenance Manual, section 3.3, C-1 Archive of Data Files).

SECTION 3: FLIGHT INSPECTION SUPPORT

This procedure provides instruction for adjusting localizer and glide slope alignments and widths during a flight inspection by a Civil Aviation Authority. It can also be used during the FTM theodolite –based Flight Inspection described in the preceding section.

3.1 Final BIT Monitor Limits Documentation

Monitor limits must be recorded prior to flight inspection. Any changes to monitor limits after flight inspection may require a partial flight inspection of signal alignment.

The “nominal” values for key performance parameter are “the values that are expected.”

1. On the MI, open the Monitor Limits windows.
2. For each Monitor Limits window, record the final BIT nominal values in Appendix B, Table B-8, Final Bit Nominals. Do NOT change any nominal values at this point.
3. Exit the Monitor Limits windows.

3.2 Reduced GTU Output during Flight Inspection

During the Standard Service Volume (SSV) flight inspection profile, the light inspector will request that the TLS be operated with the ILS uplink signals attenuated to their 3 dB alarm points. Inform the flight inspector that it will take approximately 5 minutes to set up the system for reduced output power mode, and use the following steps:

To Insert an Attenuation Condition (Change to RF Power condition):

1. Stop system software.
2. Use the FlightCheck tool to set the monitor limits 3 dB lower. Select the GTU tab, press “**Down 3 dB**” and then press **Update**.
3. Insert a 3 dB attenuator into the desired output at the rear of the GTU. Ensure all connections are tight. If two 3 dB attenuators are available then both can be installed at this time, one on the localizer and the second on the glide slope transmit cables.
4. Start system software and when BIT completes, verify that the system is in Track.

To Remove an Attenuation Condition (return to Normal):

1. Stop system software.
2. Use the FlightCheck tool to set the monitor limits 3 dB higher. Select the GTU tab, press “**Up 3 dB**” and then press **Update**.
3. Remove the 3 dB attenuator from the transmitter output at the rear of the GTU, making sure connections are all tight.
4. Start system software and after BIT completes verify that the system is in Track mode.

3.3 Adjustment of Localizer/Glide Path Signals

The objective of this step is to respond to Flight Inspection measurements of the localizer and glide slope signal and correct any deficiencies measured by the Flight Inspector.

1. Ensure all configuration files are available on the MIU by copying Base2 configuration files from “Base2\c:\run” to “\\miu\c:\ \run”. Configuration files include the following:
   - cal.dat
   - monitor_limits.dat
   - survey.dat
   - settings.dat
   - approach.dat
   - tilt.dat
2. Open the FlightCheck tool located in c:\Tools\FlightCheck on the MIU.
3. Press the “Select...” button and set the working directory to the MIU “c:\run”. This determines the location where the tool maintains its working copy of the site configuration files, and most importantly the cal.dat and approach.dat file it will be updating.
4. Select the Control tab and press “Stop System Software”.
5. Select the appropriate tab from the window as shown in Figure 3-1 and adjust the desired parameter to be within tolerance. Press the **Update** button. Note that the tool creates a sub-folder named “changesXX” and writes copies of both the original and updated files to this folder. “XX” is incremented each time “Update” is pressed. The tool then sends the modified file to all system directories.
6. Select the “Control tab” and press “Start System Software”
7. Repeat data entry for other parameters, as necessary, to support flight inspection. All FlightCheck software tool tabs are described in the table below.

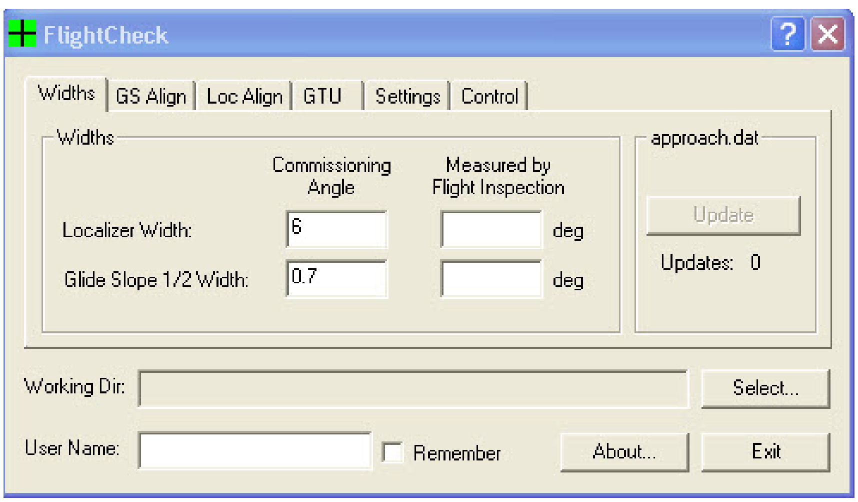


Figure 3-1 Flight Check Window

Table 3-1 Flight Check Data Entry

Tab	Field	Instructions
Widths	Localizer Width	Enter the localizer width reported by flight inspection
	Glide Slope ½ Width	Enter the glide slope ½ width reported by flight inspection
	Working dir	Always set to c:\run on the MIU
	Base1	Always select for all tabs and all adjustments
	Base2	Always select for all tabs and all adjustments
All	User Name && Remember	Enter the user name of the operator making adjustments. Select Remember to not have to reenter for each change.
GS Align	Reported Glide Slope Angle	Enter the glide path angle reported by the flight inspector. The software will automatically correct the appropriate direction
Loc Align	Loc Align Beyond 1 mile uA	If the Localizer alignment is offset by less than 20 μA, enter the Localizer alignment reported by the flight inspector in the field labeled “phase μA” and select Left or Right according to which side the error predominates. See further instructions below on the sign of the correction.
	Course Bend at Threshold uA	Enter the measured TCH
GTU	Up 3dB	Press this in coordination with removing the uplink attenuator(s)
	Down 3dB	Press this in coordination with adding the uplink attenuator(s)
	GTU1	Options for multiple GTU’s are on this page as well
Settings	GS Range	Enter the range of the glide slope signal service volume
	Loc Range	Enter the range of the Localizer signal service volume
	Record Trk RAM	Change the current setting - without having to use the settings dialogue on the Maintenance Interface (MI)
Control	Start / Stop System Software Buttons	These buttons start and stop the system without having to use the MI. Each change to one of the above parameters requires stopping and then starting the TLS base station software in order for the change to become effective. It is possible to run this Standalone using the “Run Standalone” check box.

8. In configuration case 1 below, if the signal predominates on the 90 Hz side and would cause an aircraft to steer to the right in error by an amount equal to 5μA, then the error will be called “5 right”. This also means that if the aircraft has centered needles, the aircraft will line up to the right of the extended runway centerline.

9. Determine which sign for the correction to the localizer guidance must be selected using the table below:

Configuration case	TLS ESA and ASA orientation	Flight Inspection reports:	Then enter in the flight check tool:
1		5 right	5 right
		5 left	5 left
2		5 right	5 left
		5 left	5 right

10. Next, on the TLS Flight check tool input the reported microampere measurement as 5 and select the radio button corresponding to the table above to correct the centerline of the TLS localizer guidance.

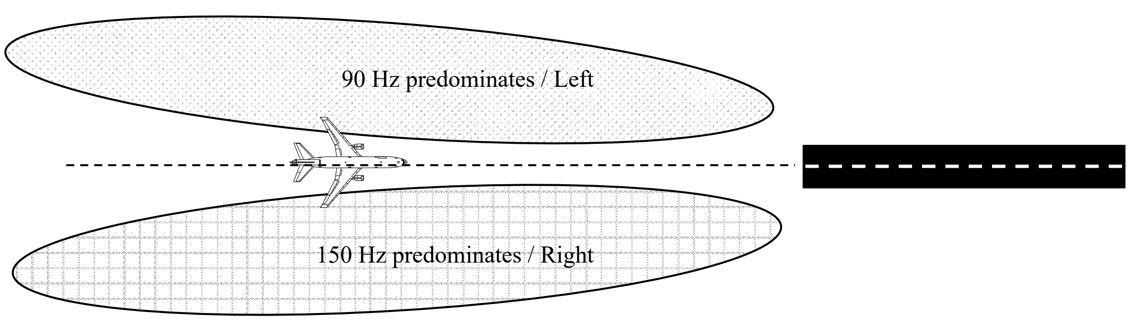


Figure 3-2 Alignment of Localizer

11. Exit the FlightCheck software after the flight inspection is complete.
12. Make a backup copy of all Base2 configuration files (*.dat file named below) in a backup folder on "\\base2\c:\run\backup". Copy these files to make a backup on the MIU computer as well on "\\miu\c:\run\backup".
    - cal.dat
    - monitor_limits.dat
    - survey.dat
    - settings.dat
    - approach.dat
    - tilt.dat
13. Record completion of Flight Inspection in Appendix C.

3.4 Verification Requirements

Table 3-2 Transponder Landing System Verification

Service	Verification Parameters	Reference Standards and Tolerance / Monitor Limits
Coverage	Power Output Receiver Sensitivity	See Section 1: Standards and Tolerances in the ANPC Maintenance Manual (document number 020-00074) for data.
Course accuracy and sensitivity	Course Accuracy and Sensitivity Antenna Alignment (ESA) Modulation Equality	
Identification	Modulation (1020 Hz and A, B, and C tones)	
Flag action	Modulation (90/150 Hz)	
Operational continuity and integrity	Monitor Shutdown Delay Remote Status Indication Remote Entry	
Monitor	RF Carrier Power Modulation (90/150 Hz) Tracking Accuracy	

Normal verification interval: Monthly.

Maximum verification interval: 45 days.

Person responsible for verification: TLS certified technician.

Verification entry in facility maintenance log: TLS verified.

Appendix A Abbreviations and Definitions

Abbreviation	Definition
µA	Micro Amperes
AGL	Above Ground Level
ANPC	Advanced Navigation & Positioning Corporation
ASA	Azimuth Sensor Assembly (previously called LAOA)
ATA	Azimuth Time-Of-Arrival (TOA) Antenna
ATC	Air Traffic Control
BIT	Built-In-Test
CAA	Civil Aviation Authority
Cal/BIT	Calibration/Built-in-Test
CNTL	Control
CPU	Central Processing Unit
dB	Decibel
DDM	Differential Depth of Modulation
ESA	Elevation Sensor Assembly (previously called GAOA)
FCC	Federal Communications Commission
FTM	Flight Test Monitor
GAOA	Glide slope Angle-of-Arrival (previous term for ESA)
GS	Glide slope
GTU	Guidance Transmitter Unit
Hz	Hertz
IAP	Instrument Approach Procedure
ID	Identification
IFR	Instrument Flight Rules
ILS	Instrument Landing System
IM	Integrity Monitor
KVM	Keyboard Video Mouse
LAOA	localizer Angle-of-Arrival (previous term for ASA)
LOC	Localizer
LRU	Line Replaceable Unit
nm	nanometer
NM	Nautical Mile
MI	Maintenance Interface
MIU	Maintenance Interface Unit
OS	Operating System
RCU	Remote Control Unit
RF	Radio Frequency
RMA	Return Material Authorization
RSA	Runway Safety Area
RSM	Remote Status Monitor (optional feature)
SSV	Standard Service Volume
TLS	Transponder Landing System

Abbreviation	Definition
TOA	Time-of-Arrival
UHF	Ultra High Frequency
UPS	Uninterruptible Power Supply
VAC	Volts Alternating Current
VFR	Visual Flight Rules
VHF	Very High Frequency

Appendix B TLS Site Validation Data Sheet

Copy the following pages and complete with the sponsor/sponsor's representative. One copy should be retained by the customer, one by the site manager, and the original to ANPC.

When completing validation testing after an LRU replacement, copy the pages of the relevant data sheet, complete only the sections required by the procedure, sign off results, incorporate into the baseline Site Validation Data Sheet, and file per documentation requirements.

Table B-1 Calibration and Validation: FTM Link Validation

Para	Test	Min	Measured	Max	Units
SECTION 2:	Calibration and Validation				
2.1.2 -8	FTM link operational validation				✓

Notes:

Table B-2 Calibration Profiles

Para	Successful Calibration Profiles	Raw File Name	Notes
2.2 1a	Level Flight - 1000' AGL		
2.2.1.2 1b	Level Flight - 1600' AGL		
2.2.1.2 1e	Glide Path Following - 1600' AGL, then following Glide Path from 5 NM		
2.2.1.2 1c	Localizer Crossing - @ 5 NM and @ 1600' AGL or 500' above minimum safe altitude		

Customer Initials _______________ ANPC Initials_______________

Para	Successful Calibration Profiles	Raw File Name	Notes

Table B-3 Flight Test Completion

Para	Test	Min	Measured	Max	Units
2.2.1.2 -5	Verify flights complete				✓
2.2.1.2	Valid Cal.dat created from flight data				✓

Table B-4 Signal Performance Verification

Para	Successful Validation Profiles	Raw File Name	Notes
Signal Performance Verification			
2.5	Localizer arc (left to right)		
	Localizer arc (right to left)		
	Localizer standard approach		
	glide slope crossing		
	glide slope standard approach #1		
	glide slope standard approach #2		
	glide slope below path approach		

Customer Initials ______________ ANPC Initials______________

Table B-5 Final BIT Nominals

Para	Test	Min	Measured	Max	Units
3.1	**Final BIT Nominals**				
	ESA Nominals				
	ESA High Phase				Cnts
	ESA Medium Phase				Cnts
	ESA Low Phase				Cnts
	ESA Average Power High				mV
	ESA Average Power Medium				mV
	ESA Average Power Low				mV
	ESA Average Power Reference				mV
	ESA TOA Bias Near				ns
	ESA Delta Phase High				Cnts
	ESA Delta Phase Medium				Cnts
	ESA Delta Phase Low				Cnts
	ESA Perpendicular Tilt				deg
	ESA Parallel Tilt				deg
	ASA Nominals				
	ASA High Phase				Cnts
	ASA Low Phase				Cnts
	ASA Average Power High				mV
	ASA Average Power Low				mV
	ASA Average Power ATA				mV
	ASA Average Power Reference				mV
	ASA TOA Bias Near ATA				ns
	ASA TOA Bias Near				ns
	ASA Delta Phase High				Cnts
	ASA Delta Phase Low				Cnts
	GTU Nominals				
	Localizer Carrier Sample Power				mV
	Localizer Reflected Power				mV
	Glide slope Carrier Sample Power				mV
	Glide slope Reflected Power				mV
	Interrogation Transmitter Nominals				
	P1/P3 Nominal Amplitude				mV

Customer Initials _______________ ANPC Initials_______________

Para	Test	Min	Measured	Max	Units
	P2 Nominal Amplitude				mV

Customer Initials _______________ ANPC Initials_______________

Appendix C ANPC Site Data Verification Sign-off

Notes / Remarks: ____

Released by: ______________________________
ANPC Installation Manager

Accepted by: ______________________________
TLS Owner / Representative

Customer Initials ______________ ANPC Initials______________

Appendix D ANPC TLS Manual Recommended Change Form

The following form is intended to document recommended changes pertaining to the Technical Instruction Manual. The TLS Manual Recommended Change Form may be legibly handwritten or typed and must include the revision level of the Manual being discussed.

The recommendation sheet is intended to be used as master copies from which to make working recommendation sheet.

Scan and email the completed form to Email CustomerSupport@anpc.com, or use standard mail to:

ANPC
489 N. 8th Street, Suite 203
HOOD RIVER, OR 97031
ATTN: Customer Support

Ph: 1-541-386-1747

TLS MANUAL RECOMMENDED CHANGE FORM

This recommended change pertains to:

- ☐ Safety
- ☐ Operation
- ☐ Maintenance
- ☐ Documentation

Recommendation: